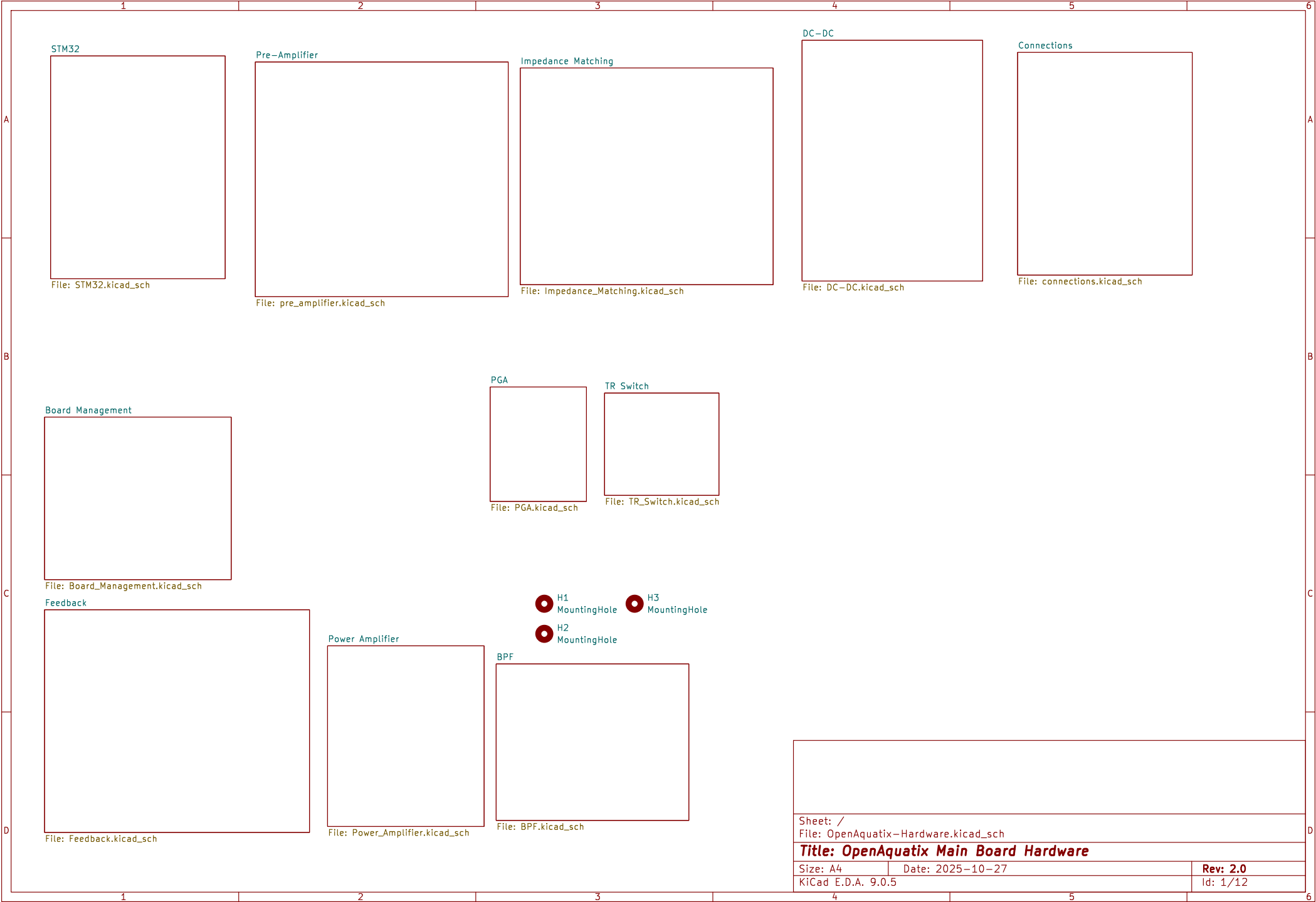
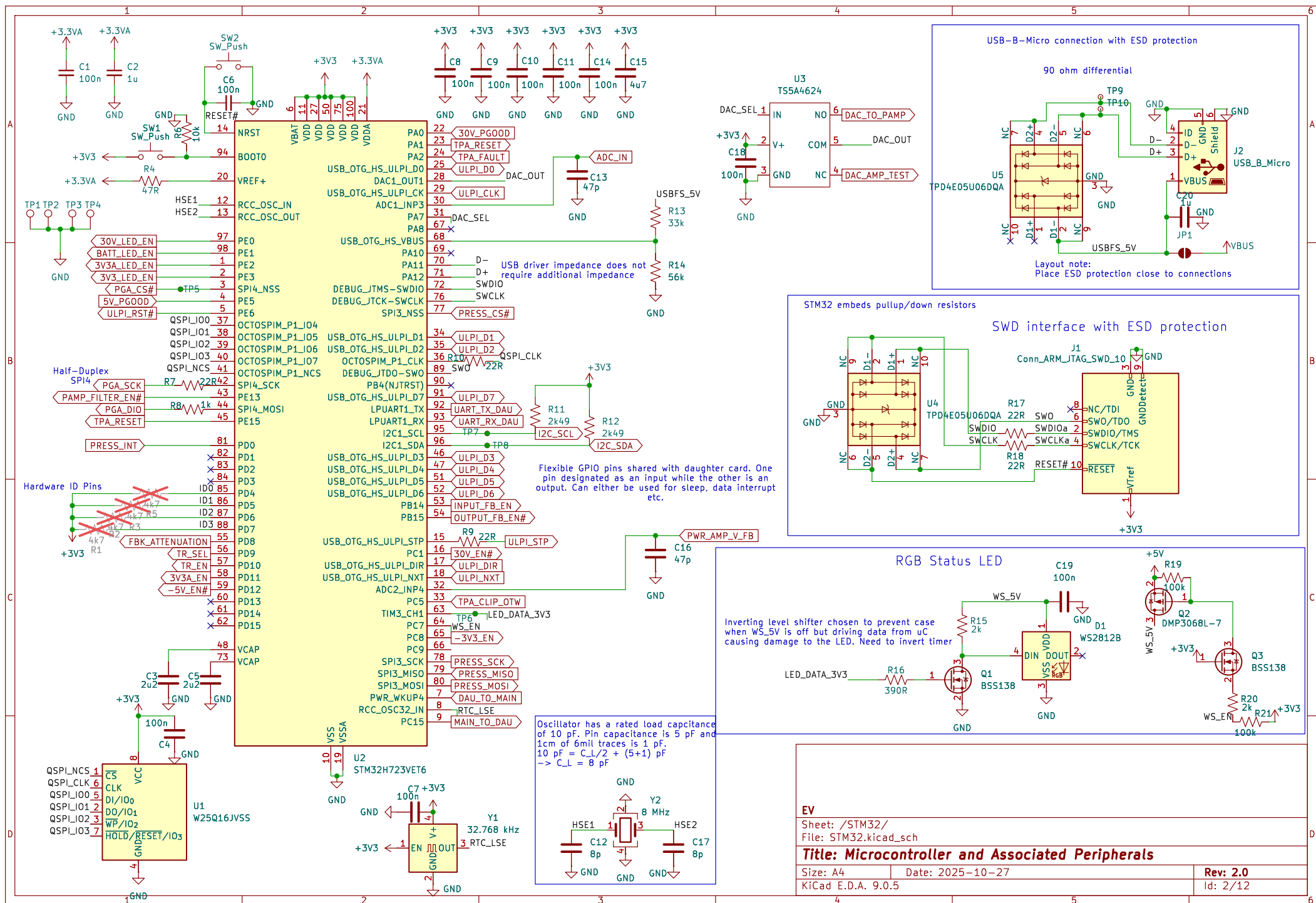
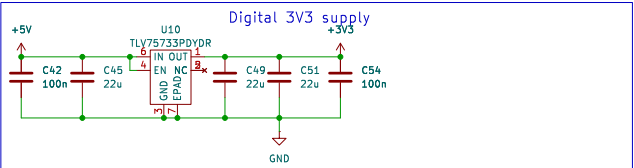
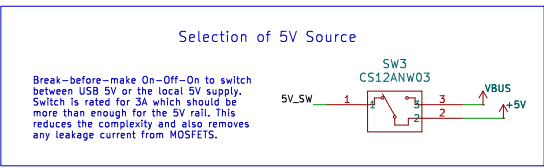
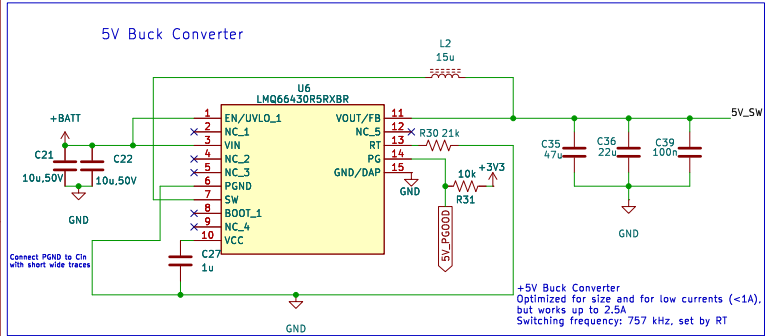
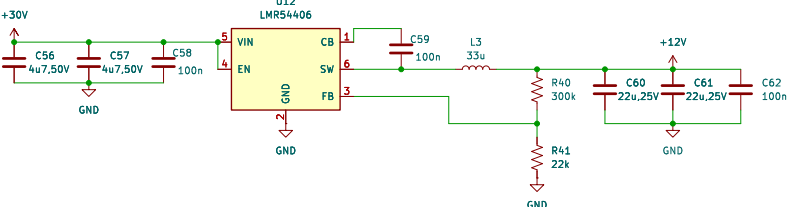
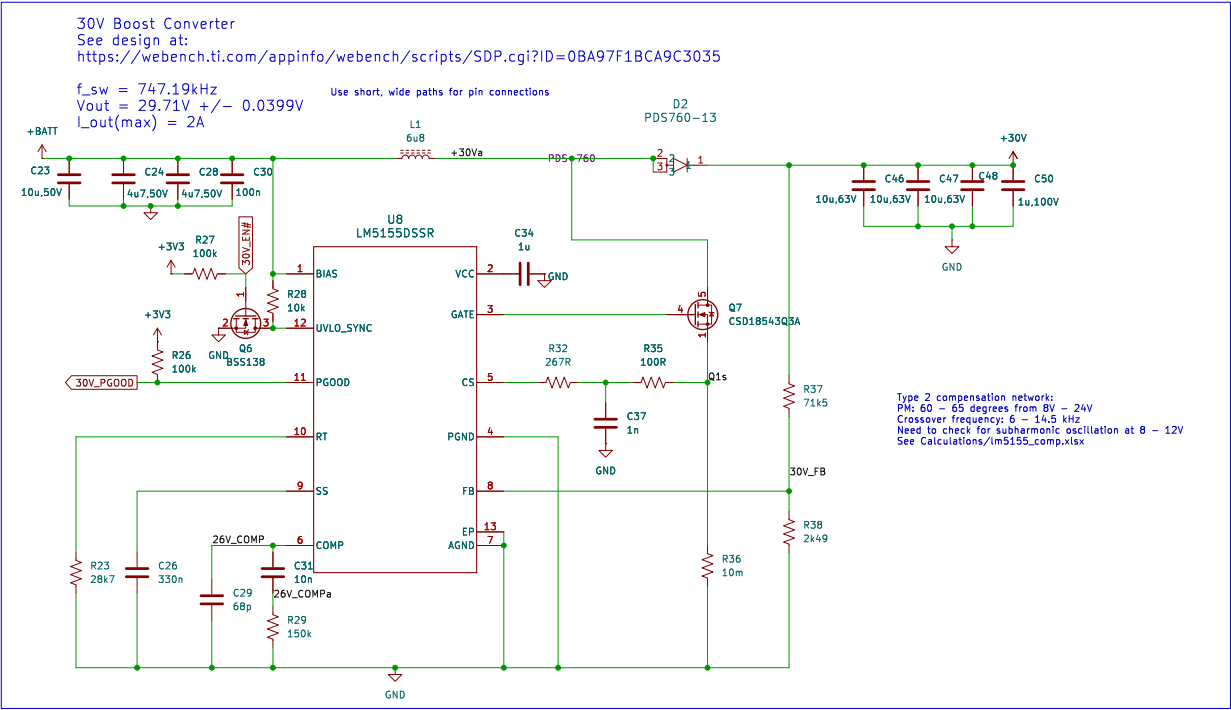
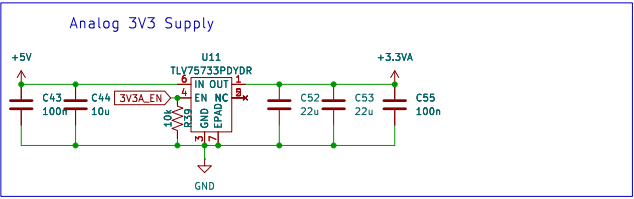
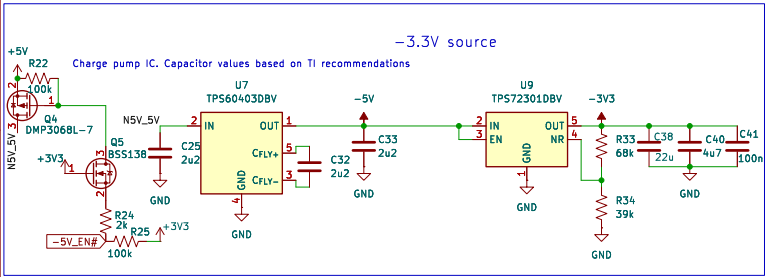




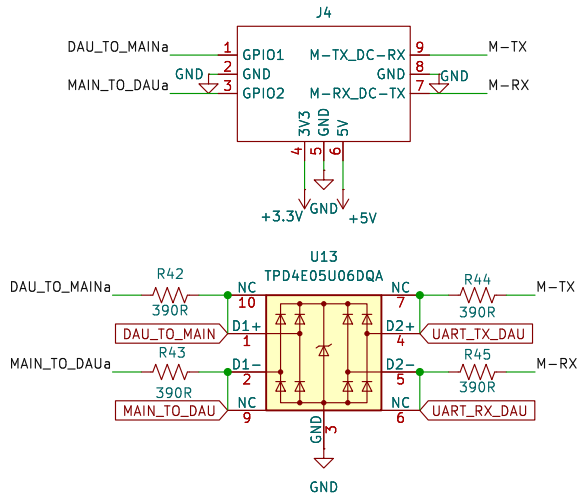




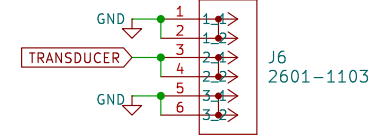
TLV75733 Specs:
 Iout: 1A
 Vout: 3.3V
 Minimum input/output capacitance: 1 uF
 Line regulation: 0.06%
 Load regulation: 2.1% over 0.1mA – 1A. Since this is a linear relation (Fig 5-13) and the analog circuit uses < 50 mA I am not worried about this impacting analog performance.
 PSRR: -40 dB @ 700 kHz, C_{out}=45 uF, Vin=5V
 I am slightly concerned about temperature performance at low temperatures, however, as the temperature will spike when transmitting which could cause fluctuations in the output voltage



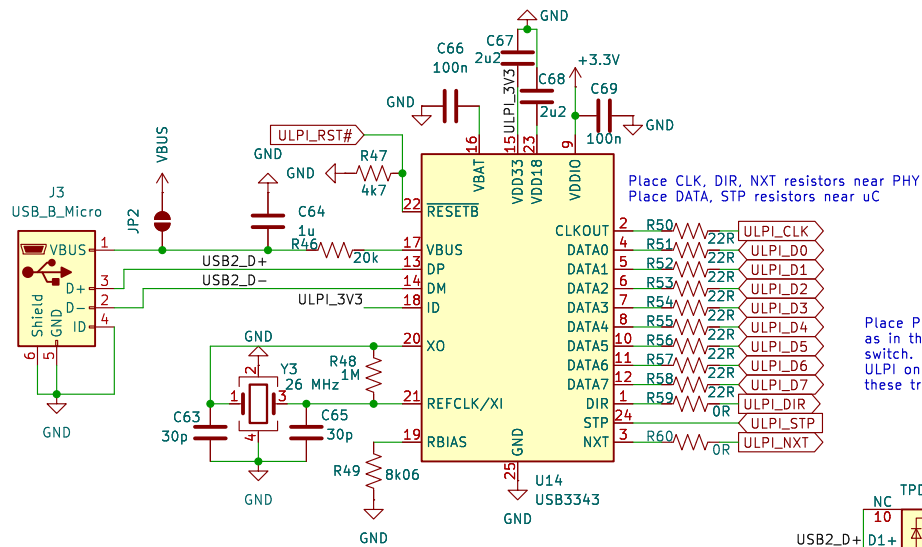
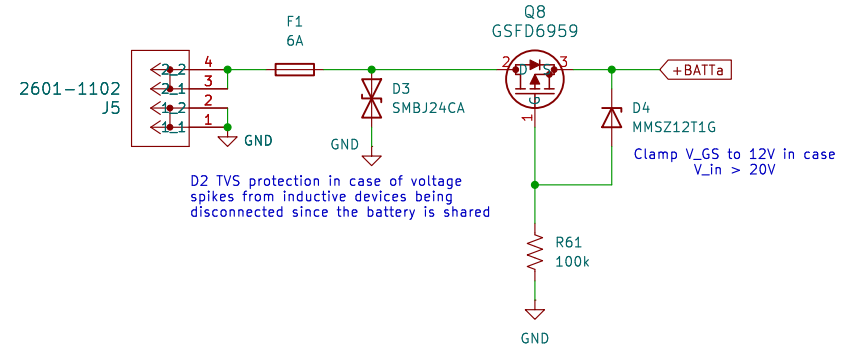
Daughter Card Connections



Hydrophone Connections



P-Channel MOSFET for RPP:
 $R_{DS(on)} = 12m\Omega$
 $V_{DS(max)} = -60V$
 $V_{GS(max)} = +/- 20V$

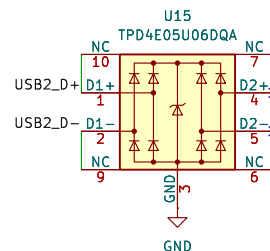


Rated load capacitance of 20 pF.
 Assuming 1cm traces and PHY
 capacitance of 4pF.

$$20 \text{ pF} = \frac{C_L}{2} + (4+1)$$

-> $C_L = 30 \text{ pF}$

Place PHY as close as possible to DIR and NXT pins
 as in the uC they are connected to a ~300 ohm analog
 switch. Forum posts mention issues with external
 ULPI on STM32H7 100-LQFP in particular. Prioritize
 these traces

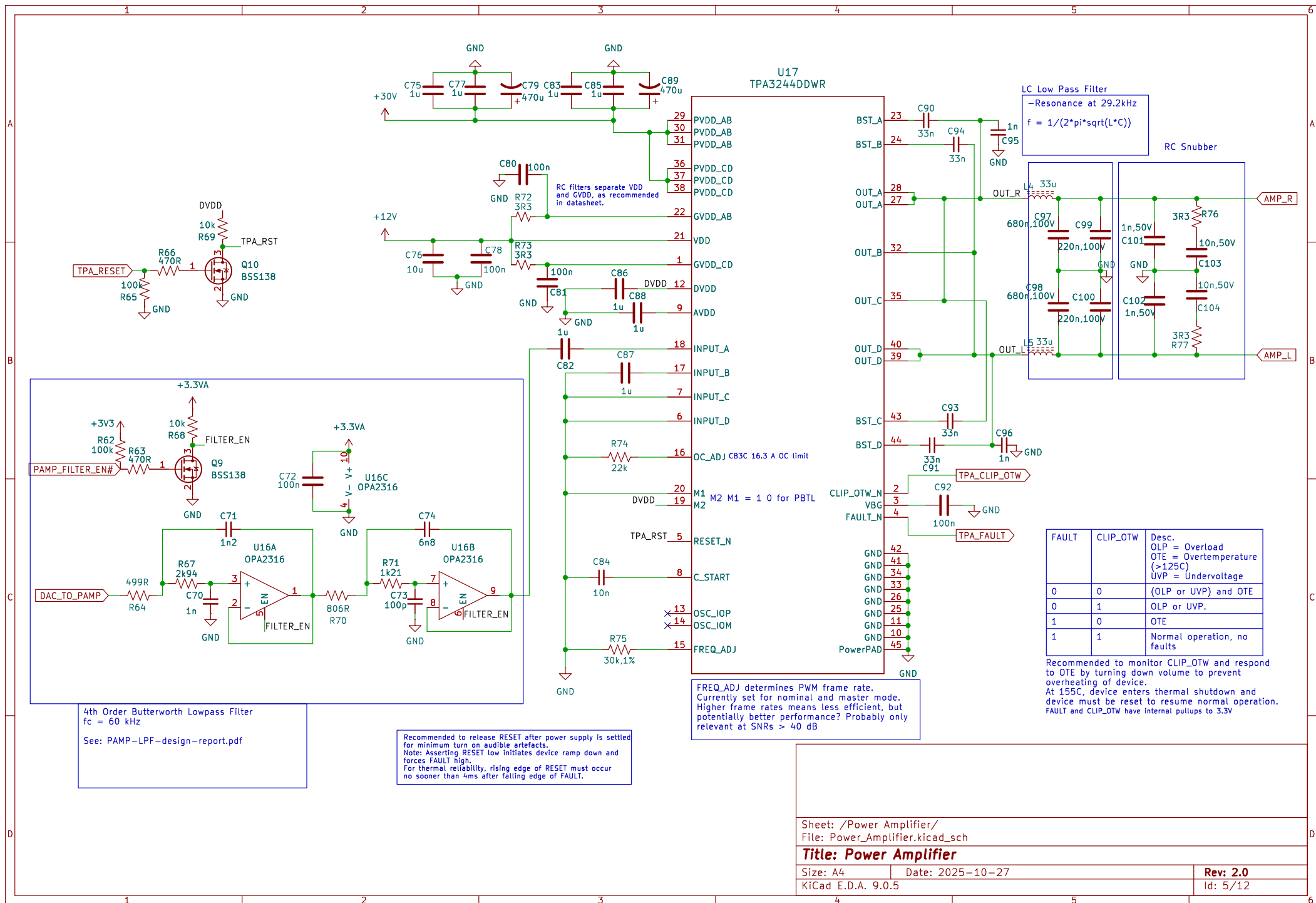


Sheet: /Connections/
 File: connections.kicad_sch

Title: External Connections to Main Board

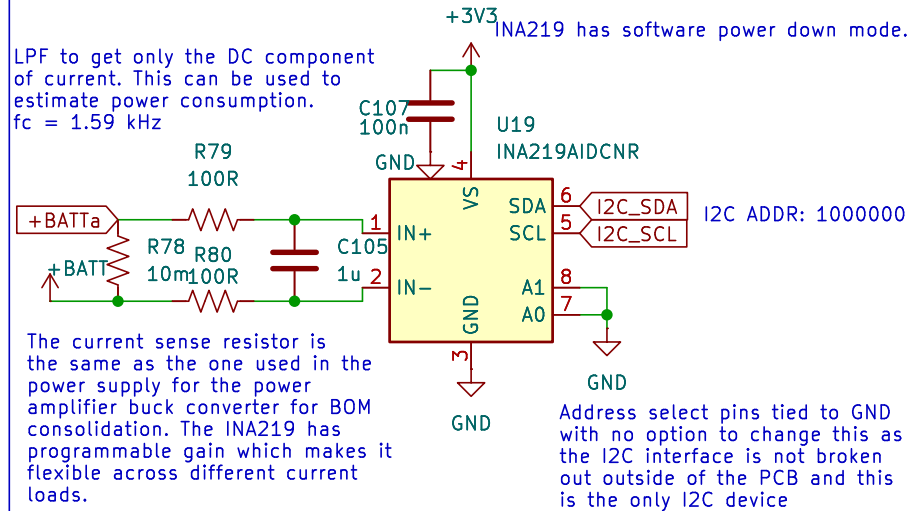
Size: A4 Date: 2025-10-27
 KiCad E.D.A. 9.0.5

Rev: 2.0
 Id: 4/12

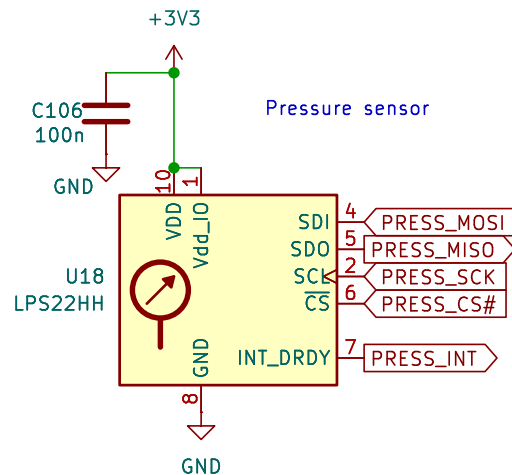
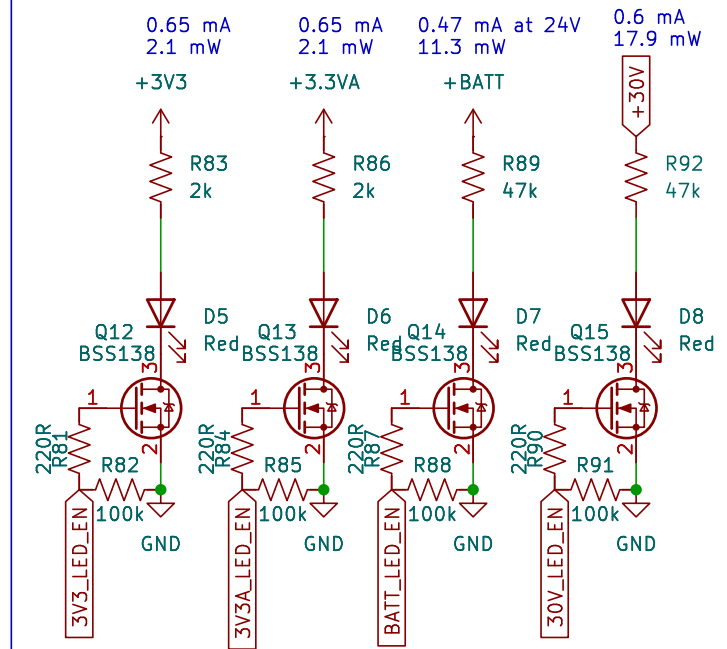


Power Consumption

Voltage and current sense IC used to detect the amount of power used by all of the electronics. Not strictly necessary for any design requirements, but useful for debugging issues and for testing future implementations in low power applications.



Power Good Indicators



Sheet: /Board Management/
File: Board_Management.kicad_sch

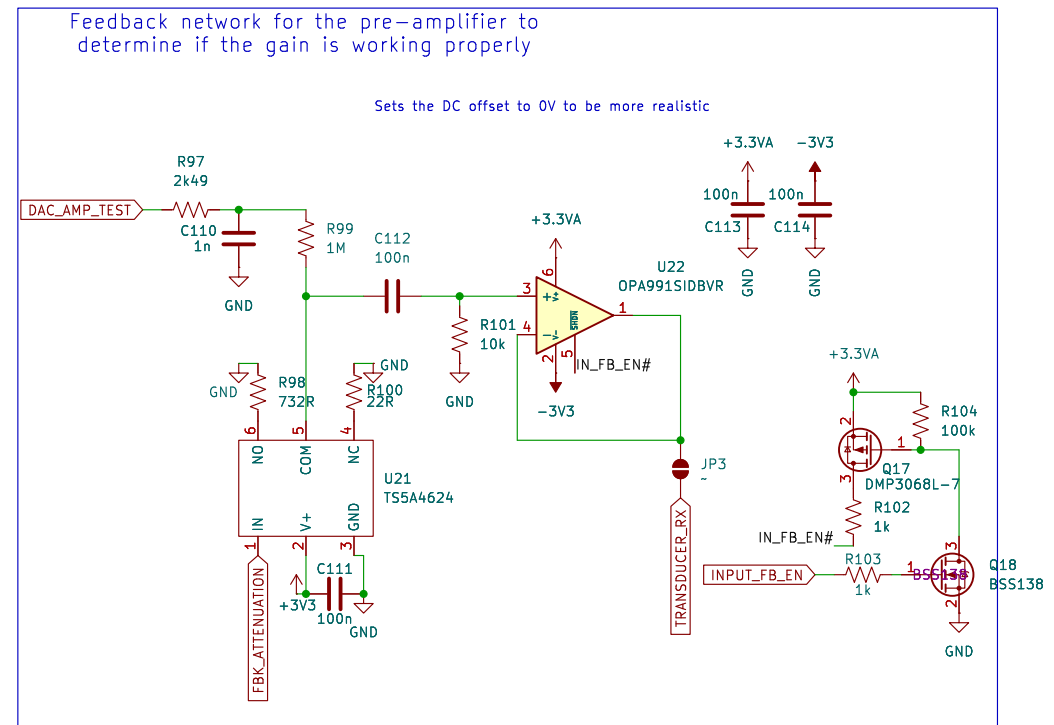
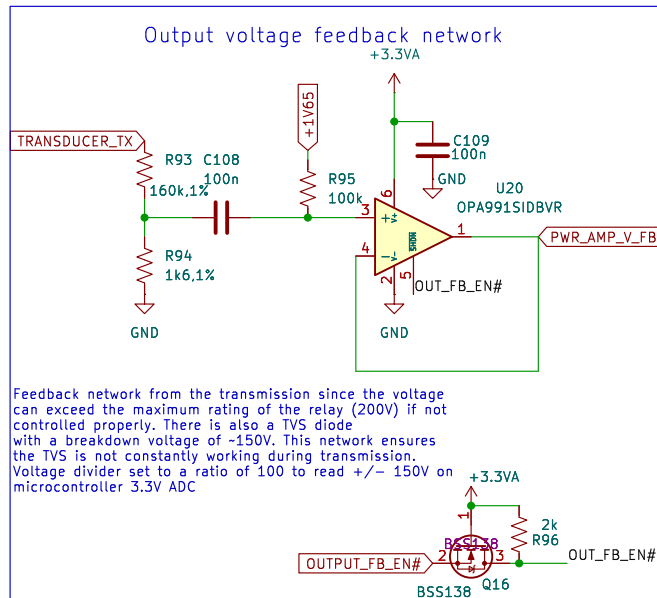
Title: Board Management

Size: A5 Date: 2025-10-27

Rev: 2.0

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Id: 6/12



Sheet: /Feedback/
File: Feedback.kicad_sch

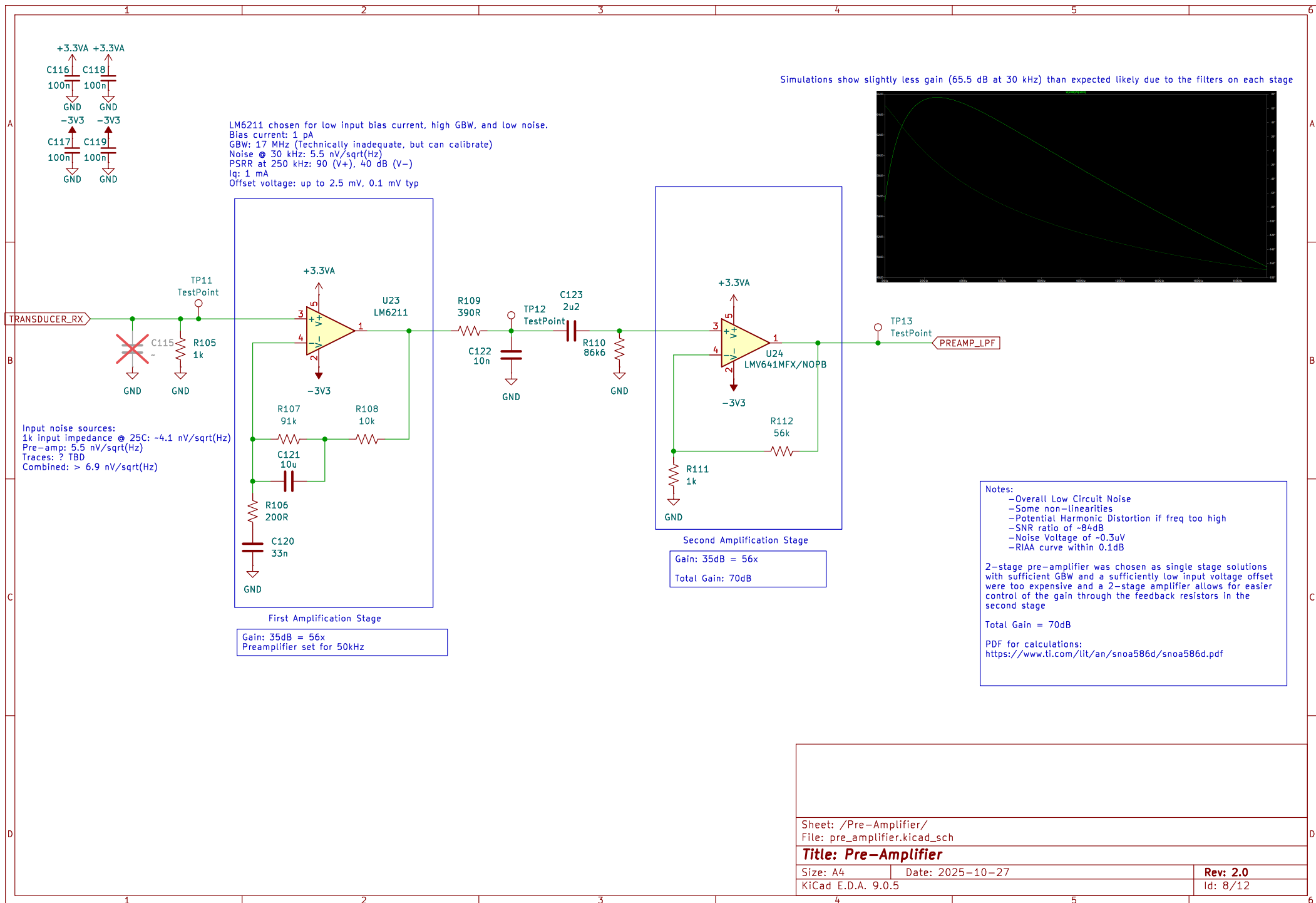
Title: Input/Output Feedback

Size: A4 Date: 2025-10-27

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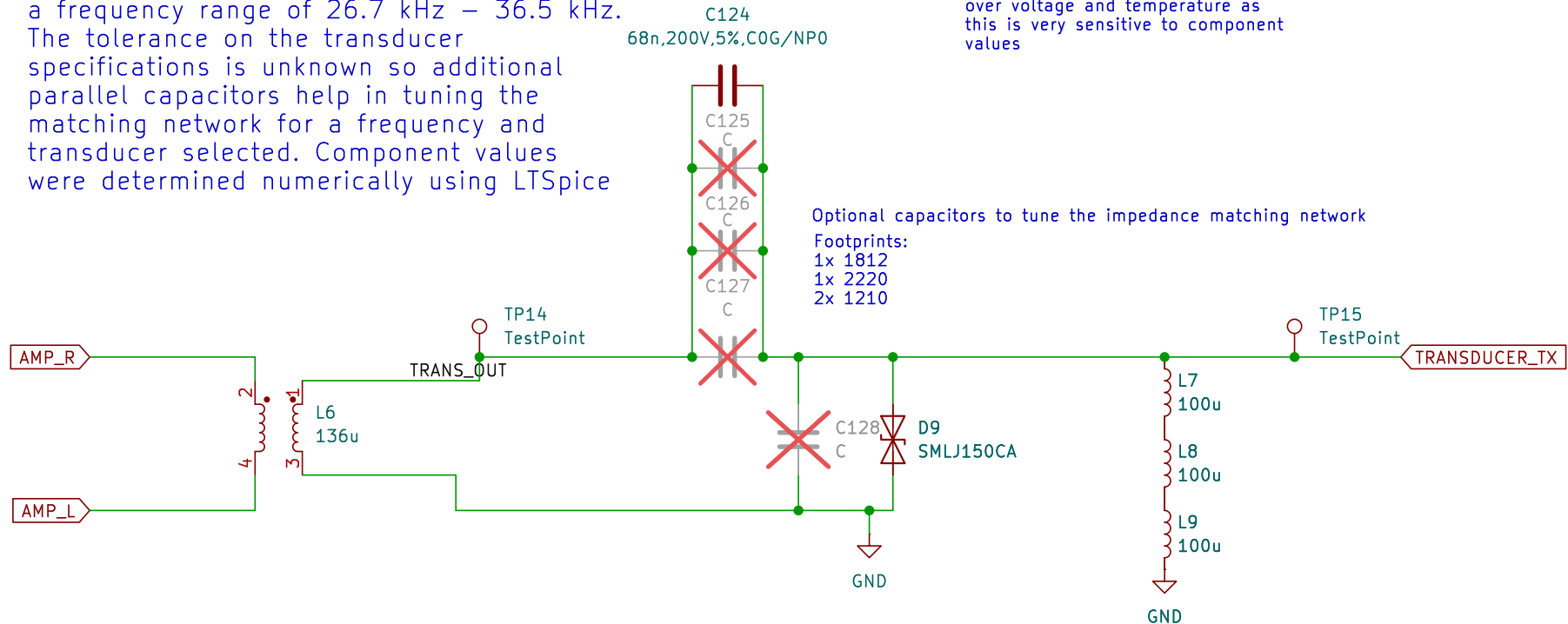
Rev: 2.0

Id: 7/12



LTSpice simulations show a power factor
> 0.5 across component tolerances and
a frequency range of 26.7 kHz – 36.5 kHz.
The tolerance on the transducer
specifications is unknown so additional
parallel capacitors help in tuning the
matching network for a frequency and
transducer selected. Component values
were determined numerically using LTSpice

Note:
Only use ceramic capacitors with a
COG/NPO dielectric for stability
over voltage and temperature as
this is very sensitive to component
values



Optional capacitors to tune the impedance matching network
Footprints:
1x 1812
1x 2220
2x 1210

Sheet: /Impedance Matching/
File: Impedance_Matching.kicad_sch

Title: Impedance Matching

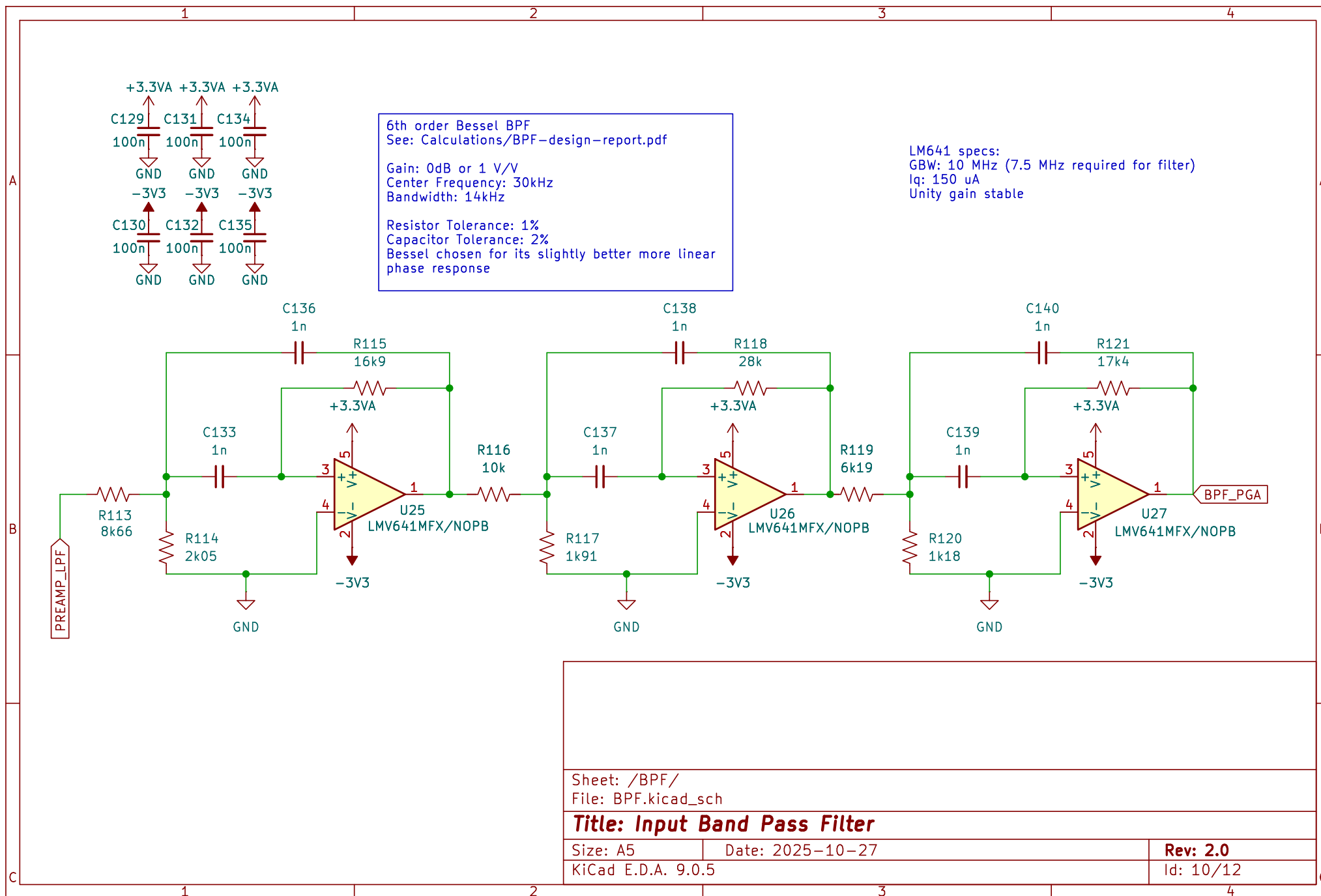
Size: A5

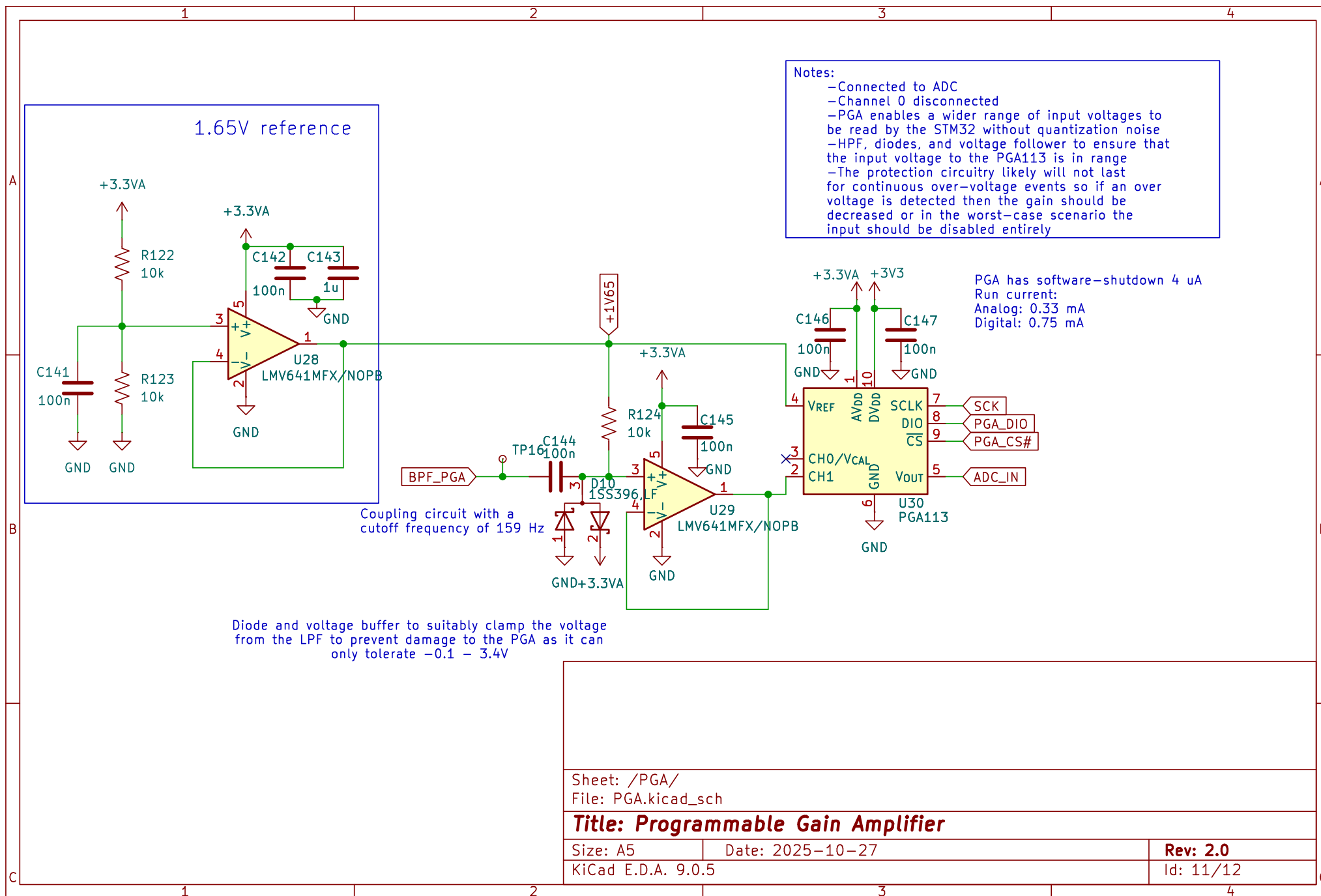
Date: 2025-10-27

Rev: 2.0

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Id: 9/12

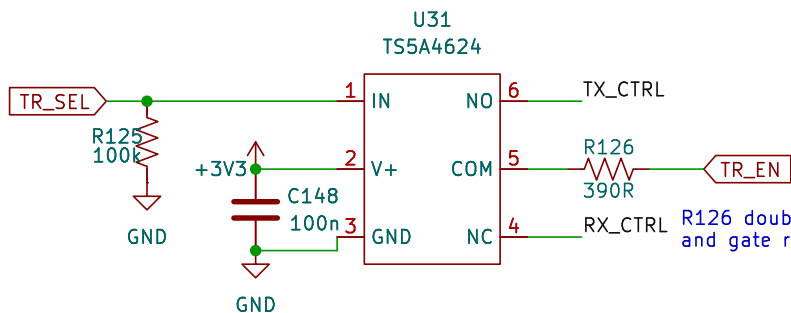




Typical relays ruled out as TR switch since they have a limited number of switching cycles, high power consumption and the fact that a low-voltage signal would need to be far away from the ground plane, significantly increasing the noise in the high impedance trace

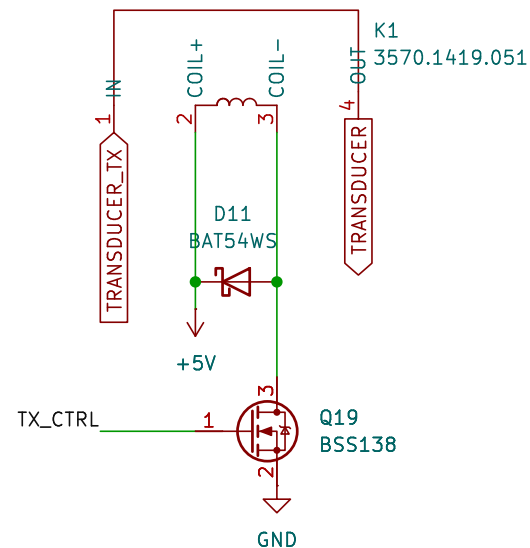
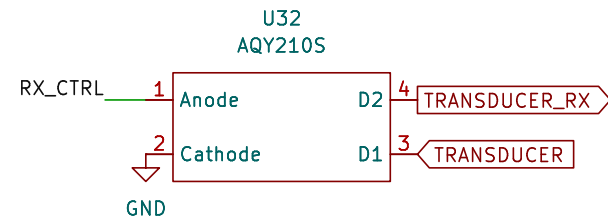
Reed relay chosen for Tx TR switch since they offer low contact resistance and lots of switching cycles while remaining acceptably low power. The open contact capacitance is high at 0.7 pF which could cause issues although unlikely. The high capacitance prevents it from being used for the Rx TR switch as too much voltage would pass through

High voltage SSR chosen for Rx TR switch since high voltage and low power which is vital for light sleep mode. The on-state resistance of 17 – 25 ohms is acceptable since the input impedance is 1k



R126 doubles as current setting resistor for PhotoMOS and gate resistor for reed relay

TR_SEL	TR_EN	
0	1	Rx ON, Tx OFF
1	1	Rx OFF, Tx ON
x	0	Rx OFF, Tx OFF



Sheet: /TR Switch/
File: TR_Switch.kicad_sch

Title: TR Switch

Size: A5 Date: 2025-10-27

Rev: 2.0

KiCad E.D.A. 9.0.5

Id: 12/12